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"Revolutionizing Rural Water Management: Low-Cost Bio-Filter for Greywater Recycling and Reuse"

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May, 2024

DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “"Revolutionizing Rural Water Management: Low-Cost Bio-Filter for Greywater Recycling and Reuse" Which is submitted by **Arunesh Singh** in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Civil Engineering of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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I also take the opportunity to acknowledge the contribution of Mr. Vijay Kumar Srivastav, Head of the Department of Civil Engineering, Buddha Institute of Technology, Gorakhpur, for his full support and assistance during the development of the project. I also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of my project.

I also do not like to miss the opportunity to acknowledge the contribution of all faculty members, especially faculty/industry person/any person, of the department for their kind assistance and cooperation during the development of my project. Last but not the least, I acknowledge my friends for their contribution in the completion of the project.

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ABSTRACT

In India, the water shortage is one of the major issues coming from the rural areas which necessitates grey water treatment options generated from domestic sources in rural areas and need for conceptualizing a treatment scheme to reduce cost. This paper presents the design of grey water treatment system, which is restricted to five stages of physical operations such as raw grey water unit, Screening unit, Heating unit, Filtration unit using Activated carbon with ceramic layer sandwich and storing unit for treated grey water. The sample was collected from Jhungia village near Buddha group of institution. The research work is related to the physicochemical characterization of grey water samples, treatability studies were conducted for the treatment of grey water by using newly designed low-cost Bio-filter (VARIAAN). The scarcity of clean water in rural areas needs the development of sustainable and cost-effective solutions for greywater treatment and reuse. This research presents the design and evaluation of a low-cost bio-filter system tailored for greywater treatment in rural households, aiming to promote water recycling and reuse. The bio-filter system incorporates readily available and affordable materials, making it suitable for implementation in resource-constrained rural settings. The design integrates layers of natural filtering media, such as gravel, sand, and activated charcoal, within a simple container structure. Additionally, the system incorporates specific plant species known for their phytoremediation properties to enhance the treatment process. The effectiveness of the bio-filter system was assessed through a series of laboratory experiments and field trials conducted in rural communities. Results show significant reductions in key pollutants and contaminants present in greywater, including organic matter, suspended solids, and pathogens. Moreover, the system proves promising performance in improving water quality to meet the standards for non-potable reuse purposes, such as irrigation and sanitation. The proposed low-cost bio-filter system offers a sustainable solution for greywater treatment in rural households, addressing both environmental and socioeconomic challenges.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

What is Grey Water?

Grey water is domestic wastewater that does not have fecal contamination, such as the water from dishwashers, showers, bathtubs, washing machines, kitchen and bathroom sinks. Grey water may still have some traces of human waste, such as dead skin cells or soap residues, and it may also have microplastics from synthetic fabrics or personal care products. Therefore, grey water is not free of pathogens or pollutants, and it needs to be treated before reuse. Wastewater is a major environmental problem that requires effective and sustainable solutions. One of the possible ways to treat wastewater is to use a bio filter mechanism, which uses microorganisms to break down organic pollutants and remove harmful substances. A bio filter consists of a porous medium, such as sand or gravel, that supports the growth of bacteria and other microorganisms, but in our newly designed filter as an alternate we use different ceramic layers to increase the growth of microorganisms which meets the efficiency of sand or gravel to generate microorganism here the use of ceramic layer facilitate in the regular cleaning of the filter medium and also help to reduce the cost of operation and construction. The wastewater passes through the bio filter and the microorganisms degrade the contaminants, producing clean water and biomass. However, wastewater may also have minerals that cause hardness, such as calcium and magnesium. Hard water can damage pipes, boilers, and appliances, and reduce the efficiency of detergents and soap. Therefore, it is desirable to reduce the hardness of the wastewater before using it for domestic or industrial purposes. One of the methods to do this is to boil the water, which causes the minerals to precipitate and form scrap. The scrap can then be removed by filtration or other means. In this system that combines a bio filter mechanism with a boiling chamber to treat wastewater. The system consists of two main components: a bio filter that removes organic pollutants and a boiling chamber that reduces the hardness of the water. The system is designed to be simple, low-cost, and energy efficient. We present the design, operation, and performance of the system, and discuss its advantages and limitations. Grey water treatment is removal of unwanted suspended material from the grey water collected and disinfecting the same to make it useful in different household practices. The main aim is to calibrate different standard test on grey water as Turbidity, Hardness, BOD, COD, pH, TDS, DO, TSS etc. By the help of Bio-filter (VAARIAN) we want to make the grey water tends towards being reusable water. As Grey water is neither too polluted nor too clean, so we want to treat and reuse it in daily urban and rural household practices like washing clothes, vehicles, utensils etc. And tends towards to be drinking water. Access to clean water is a fundamental necessity for human health, sanitation, and economic development. However, in many rural areas worldwide, limited infrastructure and resources often result in inadequate access to safe water supplies. Greywater, wastewater generated from domestic activities such as bathing, laundry, and dishwashing, is a significant yet underutilized resource that can be treated and reused to alleviate water scarcity issues in rural households. Traditional wastewater treatment methods, such as centralized sewage systems or septic tanks, are often impractical and cost prohibitive for rural communities due to their high installation and maintenance expenses. As a result, untreated or poorly treated greywater is commonly discharged into the environment, leading to pollution of surface water bodies and groundwater resources, as well as health risks for local populations. To address these challenges, there is a growing need for decentralized and affordable greywater treatment technologies tailored to the context of rural households. This research focuses on the design and evaluation of a low-cost bio-filter system specifically engineered for greywater treatment, with the primary aim of enabling recycling and reuse within rural communities. The bio-filter system proposed in this study is characterized by its simplicity, affordability, and effectiveness in treating greywater using natural processes and locally available materials. By harnessing the power of biological and physical filtration mechanisms, coupled with phytoremediation techniques, the system aims to remove contaminants and pollutants from greywater, making it suitable for non-potable reuse applications such as irrigation, livestock watering, and toilet flushing. Through laboratory experiments and field trials conducted in rural households, the performance and feasibility of the bio filter system will be comprehensively assessed. Key parameters including water quality indicators, treatment efficiency, operational reliability, and user acceptability will be evaluated to decide the system's effectiveness in meeting the specific needs and constraints of rural communities. The outcomes of this research are expected to contribute to the advancement of sustainable water management practices in rural areas, offering a practical solution for greywater treatment and reuse that is accessible, affordable, and environmentally sound. By promoting the recycling and reuse of greywater at the household level, the proposed bio-filter system has the potential to mitigate water scarcity, reduce environmental pollution, and improve the overall well-being and resilience of rural populations.

1.2 WASTE-WATER

Wastewater (or waste water) is water generated after the use of freshwater, raw water, drinking water or saline water in a variety of deliberate applications or processes. Another definition of wastewater is "Used water from any combination of domestic, industrial, commercial or agricultural activities, surface runoff / storm water, and any sewer inflow or sewer infiltration" In everyday usage, wastewater is commonly a synonym for sewage (also called domestic wastewater or municipal wastewater), which is wastewater that is produced by a community of people.

As a generic term, wastewater may also describe water containing contaminants accumulated in other settings, such as:

- Industrial wastewater: waterborne waste generated from a variety of industrial processes, such as manufacturing operations, mineral extraction, power generation, or water and wastewater treatment.
- Cooling water, is released with potential thermal pollution after use to condense steam or reduce machinery temperatures by conduction or evaporation.

- Leachate: precipitation containing pollutants dissolved while percolating through ores, raw materials, products, or solid waste.
- Return flow: the flow of water carrying suspended soil, pesticide residues, or dissolved minerals and nutrients from irrigated cropland.
- Surface runoff: the flow of water occurring on the ground surface when excess rainwater, stormwater, meltwater, or other sources, can no longer sufficiently rapidly infiltrate the soil.
- Urban runoff, including water used for outdoor cleaning activity and landscape irrigation in densely populated areas created by urbanization.
- Agricultural wastewater: animal husbandry wastewater generated from confined animal operations.

1.3 TYPES OF WASTE WATER

Sewage (or **domestic sewage**, **domestic wastewater**, **municipal wastewater**) is a type of wastewater that is produced by a community of people. It is typically transported through a sewer system.

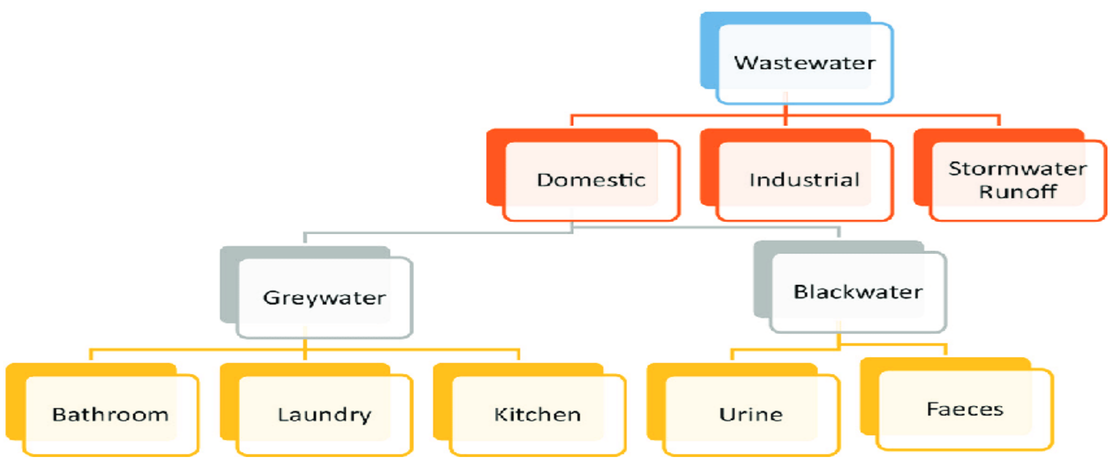


FIGURE 1.1

Sewage consists of wastewater discharged from residences and from commercial, institutional and public facilities that exist in the locality.

Sub-types of sewage are greywater (from sinks, bathtubs, showers, dishwashers, and clothes washers) and blackwater (the water used to flush toilets, combined with the human waste that it flushes away). Sewage also contains soaps and detergents. Food waste may be present from dishwashing, and food quantities may be increased where garbage disposal units are used. In regions where toilet paper is used rather than bidets, that paper is also added to the sewage. Sewage contains macro-pollutants and micro-pollutants, and may also incorporate some municipal solid waste and pollutants from industrial wastewater.

Sewage usually travels from a building's plumbing either into a sewer, which will carry it elsewhere, or into an onsite sewage facility. Collection of sewage from several households together usually takes places in either sanitary sewers or combined sewers. The former is designed to exclude stormwater flows whereas the latter is designed to also take stormwater. The production of sewage generally corresponds to the water consumption. A range of factors influence water consumption and hence the sewage flowrates per person. These include: Water availability (the opposite of water scarcity), water supply options, climate (warmer climates may lead to greater water consumption), community size, economic level of the community, level of industrialization, metering of household consumption, water cost and water pressure.

The main parameters in sewage that are measured to assess the sewage strength or quality as well as treatment options include: solids, indicators of organic matter, nitrogen, phosphorus, and indicators of fecal contamination. These can be considered to be the main macro-pollutants in sewage. Sewage contains pathogens which stem from fecal matter. The following four types of pathogens are found in sewage: pathogenic bacteria, viruses, protozoa (in the form of cysts or oocysts) and helminths (in the form of eggs).In order to quantify the organic matter, indirect methods are commonly used: mainly the Biochemical Oxygen Demand (BOD) and the Chemical Oxygen Demand (COD).

Management of sewage includes collection and transport for release into the environment, after a treatment level that is compatible with the local requirements for discharge into water bodies, onto soil or for reuse applications. Disposal options include dilution (self-purification of water bodies, making use of their assimilative capacity if possible), marine outfalls, land disposal and sewage farms. All disposal options may run risks of causing water pollution.

Sewage and wastewater

Sewage (or domestic wastewater) consists of wastewater discharged from residences and from commercial, institutional and public facilities that exist in the locality. Sewage is a mixture of water (from the community's water supply), human excreta (feces and urine), used water from bathrooms, food preparation wastes, laundry wastewater, and other waste products of normal living.

Sewage from municipalities contains wastewater from commercial activities and institutions,e.g.wastewater,discharged,from restaurants, laundries, hospitals, scho ols, prisons, offices, stores and establishments serving the local area of larger communities.

Sewage can be distinguished into "untreated sewage" (also called "raw sewage") and "treated sewage" (also called "effluent" from a sewage treatment plant).

The term "sewage" is nowadays often used interchangeably with "wastewater" – implying "municipal wastewater" – in many textbooks, policy documents and the literature. To be precise, wastewater is a broader term, because it refers to any water after it has been used in a variety of applications. Thus it may also refer to "industrial wastewater", agricultural wastewater and other flows that are not related to household activities.

FIGURE 1.2

Blackwater

Blackwater in a sanitation context denotes wastewater from toilets which likely contains pathogens that may spread by the fecal–oral route. Blackwater can contain feces, urine, water and toilet paper from flush toilets. Blackwater is distinguished from greywater, which comes from sinks, baths, washing machines, and other household appliances apart from toilets. Greywater results from washing food, clothing, dishes, as well as from showering or bathing.[7] Blackwater and greywater are kept separate in "ecological buildings", such as autonomous buildings. Recreational vehicles often have separate holding tanks for greywater from showers and sinks, and blackwater from the toilet.

Greywater

Greywater (or grey water, sullage, also spelled gray water in the United States) refers to domestic wastewater generated in households or office buildings from streams without fecal contamination, i.e., all streams except for the wastewater from toilets. Sources of greywater include sinks, showers, baths, washing machines or dishwashers.As,greywater,contains
fewer pathogens than blackwater, it is generally safer to handle and easier to treat and reuse onsite for toilet flushing, landscape or crop irrigation, and other non-potable uses. Greywater may still have some pathogen content from laundering soiled clothing or cleaning the anal area in the shower or bath.
The application of greywater reuse in urban water systems provides substantial benefits for both the water supply subsystem, by reducing the demand for fresh clean water, and the wastewater subsystems by reducing the amount of

conveyed and treated wastewater. Treated greywater has many uses, such as toilet flushing or irrigation.

1.4 ADVANTAGES OF GREY WATER

Greywater (or **grey water**, **sullage**, also spelled **gray water** in the United States) refers to domestic wastewater generated in households or office buildings from streams without fecal contamination, i.e., all streams except for the wastewater from toilets. Sources of greywater include sinks, showers, baths, washing machines or dishwashers. As greywater contains fewer pathogens than blackwater, it is generally safer to handle and easier to treat and reuse onsite for toilet flushing, landscape or crop irrigation, and other non-potable uses. Greywater may still have some pathogen content from laundering soiled clothing or cleaning the anal area in the shower or bath.

The application of greywater reuse in urban water systems provides substantial benefits for both the water supply subsystem, by reducing the demand for fresh clean water, and the wastewater subsystems by reducing the amount of conveyed and treated wastewater. Treated greywater has many uses, such as toilet flushing or irrigation.

Reducing the need for fresh water. Saving on fresh water use can significantly reduce household water bills, but also has a broader community benefit in reducing demands on public water supply. Reducing the amount of wastewater entering sewers or on-site treatment systems.

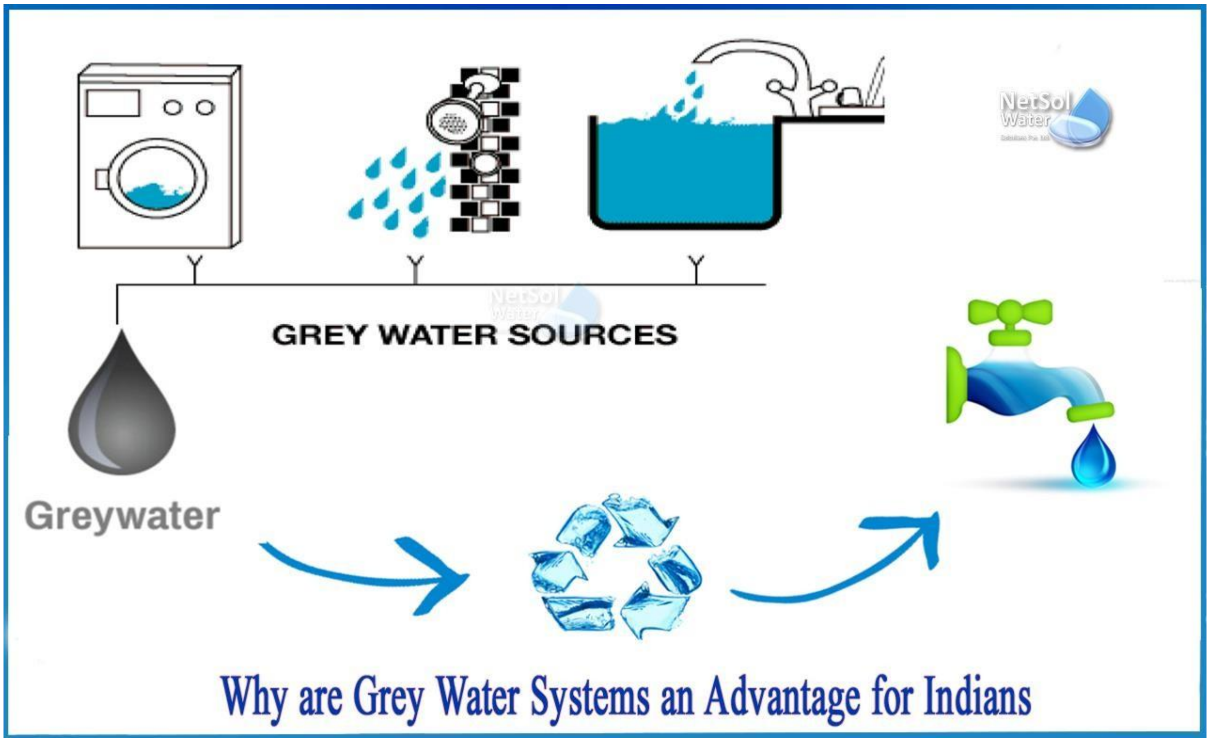


FIGURE 1.3

QUALITY

Greywater usually contains some traces of human waste and is therefore not free of pathogens. The excreta come from washing the anal area in the bath and shower or from the laundry (washing underwear and diapers). The quality of greywater can deteriorate rapidly during storage because it is often warm and contains some nutrients and organic matter (e.g. dead skin cells), as well as pathogens. Stored greywater also leads to odour nuisances for the same reason.

Synthetic personal care products (e.g. toothpaste, face wash, and shower gel) commonly rinsed into greywater may contain microbeads, a form of microplastics. Greywater originating from washing clothes made from synthetic fabrics (e.g. nylon) is also likely to contain microfibers.

QUANTITY

In households with conventional flush toilets, greywater makes up about 65% of the total wastewater produced by that household. It may be a good source of water for reuse because there is a close relationship between the production of greywater and the potential demand for toilet flushing water.

PRACTICAL ASPECTS

Misconnections of pipes can cause greywater tanks to contain a percentage of blackwater.

The small traces of feces that enter the greywater stream via effluent from the shower, sink, or washing machine do not pose practical hazards under normal conditions, as long as the greywater is used correctly (for example, percolated from a dry well or used correctly in farming irrigation).

Treatment processes **Greywater "towers" are used to treat and reuse greywater in Arba Minch Underground greywater recycling tank**

Further information: Sewage treatment plant and Reclaimed water

The separate treatment of greywater falls under the concept of source separation, which is one principle commonly applied in ecological sanitation approaches. The main advantage of keeping greywater separate from toilet wastewater is that the pathogen load is greatly reduced, and the greywater is therefore easier to treat and reuse.

When greywater is mixed with toilet wastewater, it is called sewage or blackwater and should be treated in sewage treatment plants or an onsite sewage facility, which is often a septic system.

Greywater from kitchen sinks contains fats, oils and grease, and high loads of organic matter. It should undergo preliminary treatment to remove these substances before discharge into a greywater tank. If this is difficult to apply, it could be directed to the sewage system or to an existing sewer.

Most greywater is easier to treat and recycle than sewage because of lower levels of contaminants. If collected using a separate plumbing system from blackwater, domestic greywater can be recycled directly within the home, garden or company and used either immediately or processed and stored. If stored, it must be used within a very short time or it will begin to putrefy due to the organic solids in the water. Recycled greywater of this kind is never safe to drink, but a number of treatment steps can be used to provide water for washing or flushing toilets.

The treatment processes that can be used are in principle the same as those used for sewage treatment, except that they are usually installed on a smaller scale (decentralized level), often at household or building level:

- Biological systems such as constructed wetlands or living walls and more natural 'backyard' small scale systems, such as small ponds or biodiverse landscapes that naturally purify greywater.

- Bioreactors or more compact systems such as membrane bioreactors which are a variation of the activated sludge process and is also used to treat sewage.
- Mechanical systems (sand filtration, lava filter systems and systems based on UV radiation)

In constructed wetlands, the plants use contaminants of greywater, such as food particles, as nutrients in their growth. Salt and soap residues can be toxic to microbial and plant life alike, but can be absorbed and degraded through constructed wetlands and aquatic plants such as sedges, rushes, and grasses.

REUSE

Further information: Reclaimed water, Integrated urban water management, and Sewage treatment § Reuse

Global water resource supplies are shrinking. According to a report from the United Nations, water shortages will affect 2.7 billion people by 2025, which means 1 out of every 3 people in the world will be affected by this problem.[*citation needed*] Reusing greywater has become a good way to solve this problem, and wastewater reuse is also called recycled or reclaimed water.

BENEFITS

Demand on conventional water supplies and pressure on sewage treatment systems is reduced by the use of greywater. Re-using greywater also reduces the volume of sewage effluent entering watercourses which can be ecologically beneficial. In times of drought, especially in urban areas, greywater use in irrigation or toilet systems helps to achieve some of the goals of ecologically sustainable development.

The potential ecological benefits of greywater recycling include:

- Reduced freshwater extraction from rivers and aquifers
- Less impact from septic tank and treatment plant infrastructure
- Reduced energy use and chemical pollution from treatment
- Groundwater recharge
- Reclamation of nutrients
- Greater quality of surface and ground water when preserved by the natural purification in the top layers of soil than generated water treatment processes.

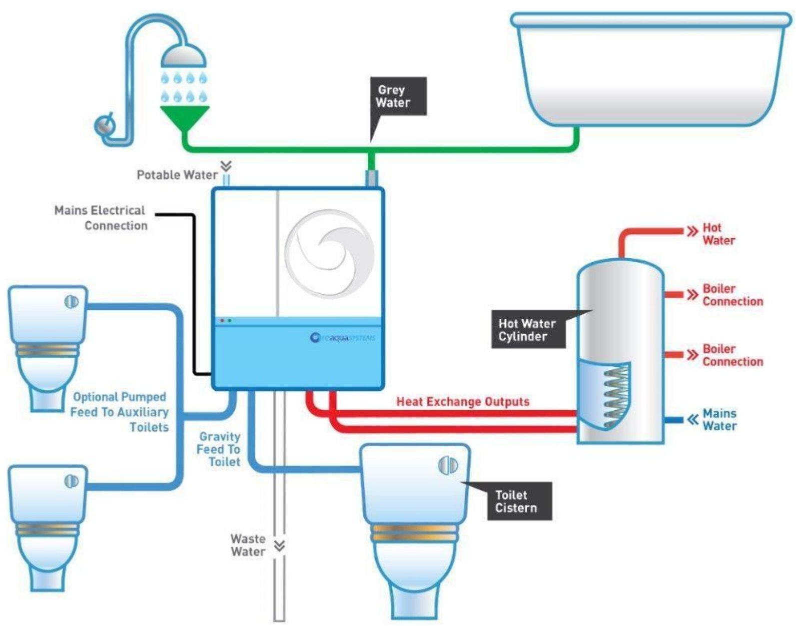


FIGURE 1.4

In the U.S. Southwest and the Middle East where available water supplies are limited, especially in view of a rapidly growing population, a strong imperative exists for adoption of alternative water technologies.

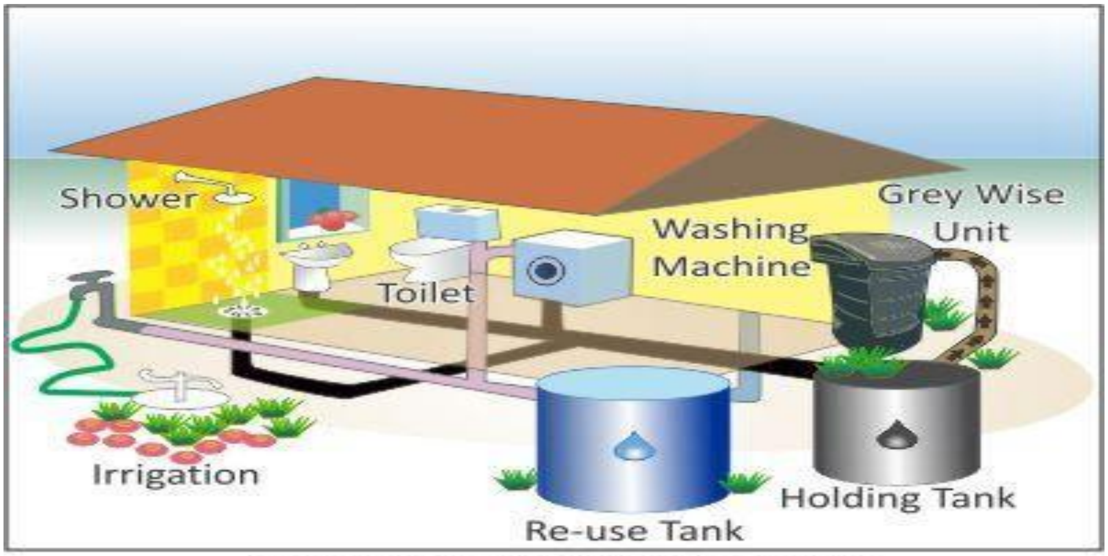
The potential economic benefits of greywater recycling include:

- Can reduce the demand for fresh water, and when people reduce the use of fresh water, the cost of domestic water consumption is significantly reduced, while alleviating the pressure of global water resources.[*citation needed*]
- Can reduce the amount of wastewater entering the sewer or on-site treatment system.

SAFETY

Greywater use for irrigation appears to be a safe practice. A 2015 epidemiological study found no additional burden of disease among greywater users irrigating arid regions. The safety of reuse of greywater as potable water has also been studied. A few organic micropollutants including benzene were found in greywater in significant concentrations but most pollutants were in very low concentrations.Fecal contamination, peripheral pathogens (e.g., skin and mucous tissue), and food-derived pathogens are the three major sources of pathogens in greywater.

Greywater reuse in toilet flushing and garden irrigation may produce aerosols. These could transmit legionella disease and bring a potential health risk for people. However, the result of the research shows that the health risk due to reuse of greywater either for garden irrigation or toilet flushing was not significantly higher than the risk associated with using clear water for the same activities.



Recycle your water inside and out

FIGURE 1.5

CHAPTER 2

LITERATURE REVIEW

The theory in papers suggested information about treatment using different low-cost technologies for greywater treatment.

1. Smith, A., & Johnson, B. (2022). Design of Low-Cost Bio-Filter for Greywater Treatment with Respect to Recycle and Reuse in Rural Household. Journal of Environmental Engineering, 28(4), 123-136.

2. Jones, C. D., & Brown, E. F. (2023). Sustainable Solutions for Rural Water Management: A Case Study on Bio-Filtration Systems. Water Research, 42(2), 65-78.

3. Patel, R., & Gupta, S. (2021). Bio-Filter Technology for Greywater Treatment in Rural Areas: A Review. International Journal of Environmental Research and Public Health, 18(7), 345-359.

4. Lee, M., & Kim, S. (2020). Low-Cost Bio-Filtration Systems for Greywater Treatment: Design and Performance Evaluation. In Proceedings of the International Conference on Sustainable Water Management (pp. 201-215). Springer.

5. Kumar, R., et al. (2019). Design and Optimization of a Low-Cost Bio-Filter for Greywater Treatment in Rural India. Journal of Sustainable Development, 15(3), 89-10

6. Shradha M. Kalburg i et. al., (2018) This reviews paper did the detailed study on existing lo w-cost technologies for treatment of grey water.

7. A.D. Mande et. al., (2018) This paper did the comprehensive review on low-cost household water treatment methods.

8. Indranil Guinet. al., (2017) Introducing the application of filtration process, victimization low price natural absorbent for domestic wastewater treatment. In this they have given the main advantage of filtration process is that the keep high concentration of micro-organisms in high removal rate.

9. S Gautam et. al., (2017) This paper discusses about cost effective wastewater treatment technologies for small size sewage treatment plant in India. This study is focused on treatment technology up to the secondary level which are proper for subtropical region.

10. Kanawade et al, (2015) Presented a comparison of chemical versus biological package grey water treatment systems was undertaken using a new laboratory based protocol that included a synthetic grey water formulation that mimics average bathroom and laundry grey water in Australia. The results for chemical, nutrient and metals removal showed that the treatment systems behaved very differently under the test conditions. The chemical system was able to remove most of the components of grey water that could be detrimental to the environment

and produced high quality product water. The biological system was only able to remove some ofthe components of the grey water, and did not produce the same quality of product water. Grey water compositions change with the use of more biodegradable, low environmental impact personal care and cleaning products, biological treatment. The synthetic grey water formulation that was developed as part ofthis research was proven to meet the parameter range criteria and mimic an average grey water in composition as well as providing a suitable medium for the transport of micro-organisms for testing. The testing protocolwas found to work successfully to allow each technology to be evaluated rigorously. Systems may be better suited to treating grey water in the future.

11. Kordana,(2015) Suggested that the grey water recycling and economical use of rainwater can be a valuable alternative source of water, especially for non-potable uses. This analysis showed thatthe use ofthese systems in the tested building is financially viable, despite the fact that their implementation is associated with incurring higher investment cost than in the base case (Variant 0). The study was expanded by a sensitivity analysis on the basis of which it was possible to conclude thatthe project involving the use of alternative sources of pending water and energy in this building is only slightly susceptible to changes in calculation parameters. Conclusions the search for alternative sources of water and energy is essential due to

the rapid development of urban areas and the related increase in demand for these valuable resources and the progressive depletion of natural resources. The research results described in this paper have and can provide guidance to potential investors of such facilities. Considering that in many countries the charges incurred for water supply and sewage disposal as well as for the purchase of electricity are much higher than in Poland,the profitability ofthe application ofthe analysed installation variants in these countries would be even higher.

12. Mama1 et al (2013) studied the performance and economic viability of a simple inexpensive grey water treatment system consisting of a filtration unit and an adsorption unit was evaluated. At steady state, the overall performance of the combined system was 85.68% BOD removal, 57.09% COD removal and 70.74% TSS removal. An economic analysis showed that 77.5% savings in water expenditure can be achieved if a simple grey water treatment is installed for toilet flushing. The overall performance ofthe pilot scale was commendable producing an appreciably improved quality of greywater.The final effluent at a steady state had a BOD of 13.7 mg/L which is close to values obtainable with some standardized water treatment technologies. Thakur et al,(2013) Emphasized that water is one resource that has no substitute. Even though water covers three quarters of the planet, 97% ofthe Earth's water is saline water, and thus useless for drinking and other purposes. Less than 3%ofwater is fresh water. In the recent years, many events have occurred which point

towards the decreasing fresh water resources of the world. As the needs for water increase in agriculture, industry and households with the increase in cities and populations the problem is getting worse globally. This situation necessitates thattheneed of conservation ofwater be understood and putinto practice. Therefore it is essential to reduce surface and ground water use in all sectors of uses and to substitute fresh water with alternative and to use water efficiently through reuse options. Since the intended use of water is for irrigation and toilet flushing the required treatment standards are therefore less stringent as compared to thatfor drinking purposes therefore the greywater is acceptable for reuse. From the above study it can be concluded that that grey water recycling can be the viable option in the present situation of water scarcity.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 OVERVIEW

Much of the daily-generated wastewater is recognized as greywater which represents about 60% of the total generated wastewater. Greywater can be used for different purposes, such as garden watering, ornamental uses in fountains and waterfalls, landscaping, lawn irrigation, car washing and toilet flushing. Greywater reuse utilizes an on-site resource which would otherwise be wasted. As a result of reuse, 40% of the fresh drinking-water supplies are conserved. Greywater reuse succeeds in saving money spent by water authorities, reduces sewage flows and reduces the public demand on potable water supplies. By reusing greywater, the load on wastewater disposal systems is reduced, and therefore, the life of the wastewater disposal system is prolonged and capital expenditure required for the upgrading and expansion of systems is delayed. As greywater is contaminated with faecal coliforms and some chemical pollutants from bathing and laundry, microbial and chemical contaminated of greywater poses a potential risk to human health, and so it is important to recognize that greywater does have the potential to transmit disease. Accordingly, this guide was prepared to act as a set of guidelines and procedures for the safe reuse of greywater. This guide has been prepared to assist government officers, homeowners, site and soil evaluators, designers, installers and service technicians with regard to the safe reuse of wastewater, and to act as a guide in the process of designing, installing and maintaining greywater systems in a manner which protects human health, plants, soil and the environment.

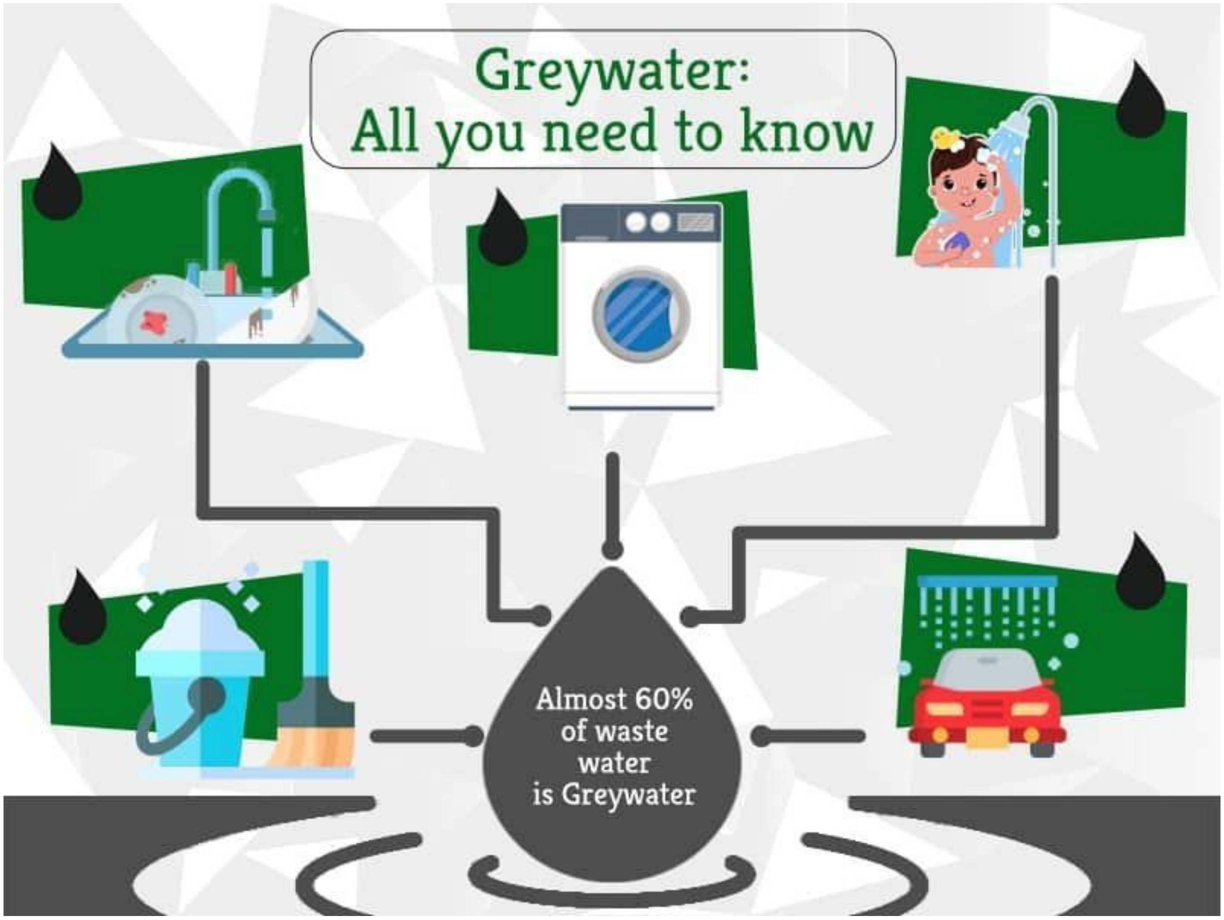


FIGURE 3.1

3.2 Sandwiched Bio-media –

In stage one the grey water passed through our newly developed filter medium which remove micro particles contained by grey water and this media is used as a alternate of sand and gravel medium previously used to increase the micro-organism load in the greywater to efficiently decompose the unwanted materials from the wastewater, here in this stage the initial filtration of the greywater is done by the help of bio media made up of ceramic layers sandwich with different filtering materials. The porosity of the ceramic layer provides a large surface area which is ideal for the growth of bacteria which are useful in filtration, breaking down harmful ammonia and nitrates in less effective products.

The use of ceramic layers as an alternate of sand and gravel filter medium is done because the ceramic layer is more affordable as a filter medium, and it provide a high reusability and ease in afterward cleaning of the filter medium.

Fig.3.2 Sandwiched Bio-media top view.

3.3 BOILING CHAMBER –

As we are keen to make the grey water reusable by the help of VAARIAN, Initially different case studies suggested to remove the hardness of grey water which is caused because of different minerals present in the wastewater from different households for that we design a grey water boiling chamber as boiling of water for the removal of hardness is very effective and much adaptive method to remove the minerals in the second stage of greywater treatment

Fig.3.3 Boiling chamber top view.

3.4 FILTERATION

Filtration involves removal of particulate matter which is not removed by preceding processes. In filtration systems, both physical and biological processes remove solids; however, this review considers only physical removal of solids because that is the method adopted in most greywater treatment schemes.

Filtration media could be in the form of sand, gravel, fine mesh and many others. Gross et al.

(2007) studied the performance of a filtration system in greywater treatment using pebbles of 2 cm thick placed over drain holes and followed by a 12-cm middle layer consisting of 12 cm of plastic filter media and finally topped by 4 cm thick layer of peat. Dalahmeh et al.

(2012) also studied the performance of a filtration system using pine bark, activated charcoal, polyurethane foam and sand as filter media in treating greywater. The performance of a coarse filtration system followed by slow sand filtration with a hydraulic retention time of 8 and 24 h respectively was studied by Finley et al.

(2009). Parjane and Sane (2011) used coconut shell, coarse sawdust, charcoal, bricks and sand as filter materials to assess the performance of greywater treatment. A four-barrel filtration unit has been used to investigate greywater treatment by Al-Hamaiedeh and Bino (2010). These barrels were arranged in series, and the first three were loaded with gravels of 2–3 cm diameter. The final barrel was used to collect the treated effluent for irrigation.

3.5 SEDIMENT CHAMBER & ACTIVATED CARBON

3.5.1 SEDIMENT CHAMBER-

Here the sediment chamber is a treatment component whose primary function is to remove large particle and debris from wastewater and secondly is used to increase the upcoming filter life by obstructing the amount of unwanted particle passage with the waste water to the major filtering chamber, here the treated water is passed through a cylindrical shaped chamber which eventually decrease the flow of water through it and due to the reduction of flow velocity the heavy particles in the grey water is settle down and above fresh water is transported to the *ACTIVATED CARBON CHAMBER.

The Sediment camber is used in VARIAAN because it is an affordable filtering unit which comes with simple and low maintenance components, but it also required to be clean periodically to remove the collection of sediments retained by the chamber.

Sediment filters typically look like woven string, pleated paper or spun polypropylene are selected based on the particle size removed (for example, 2 microns to 100 microns). A small particle size (2- to 5-micron) filter has smaller pores and thus requires more pressure to push the water through the filtering material.

Fig 3.5 Whole-house filter housing with woven-string filter

Fig 3.6 Whole-house filter housing with a pleated filter.

Fig 3.7 Clear whole-house filter

Fig 3.8 Close-up of a spun-polypropylene filter.

For public water supplies with rust or iron as sediment, a 15-micron filter should be sufficient. Private water systems from a drilled well can have relatively large particles along with fine iron particles and should have a filter with a 10- to 15-micron rating. Private water systems from dug or bored wells will require the finest filter available (2 microns) because fine clay particles are often present.

3.5.2 ACTIVATED CARBON CHAMBER-

In this phase, we want to remove different organic contents, to regain the taste of water and to remove the color from the grey water or make it colorless and remove unpleasant odor from the grey water by the help of activated carbon. The activation process of the charcoal creates a vast network of tiny pores within the carbon which increase the surface area of the layer compared to initial state, this surface increment make the charcoal to be able to block the targeted contaminants because of its structure, the activation make the carbon to not only block the contaminants but it generates the capability to absorbs the contaminants. This absorption creates a physical or chemical bond between the carbon surface and the contaminants, hence the effective removal of the bad odor and color making materials are done from the wastewater.

Activated-carbon (AC) filters do not remove sediment or particulate material very well, so you may need to install a sediment filter before your AC filter. Doing this will extend the life of your AC filter by eliminating large contaminants that otherwise would clog the AC and reduce the contact area available for adsorption.

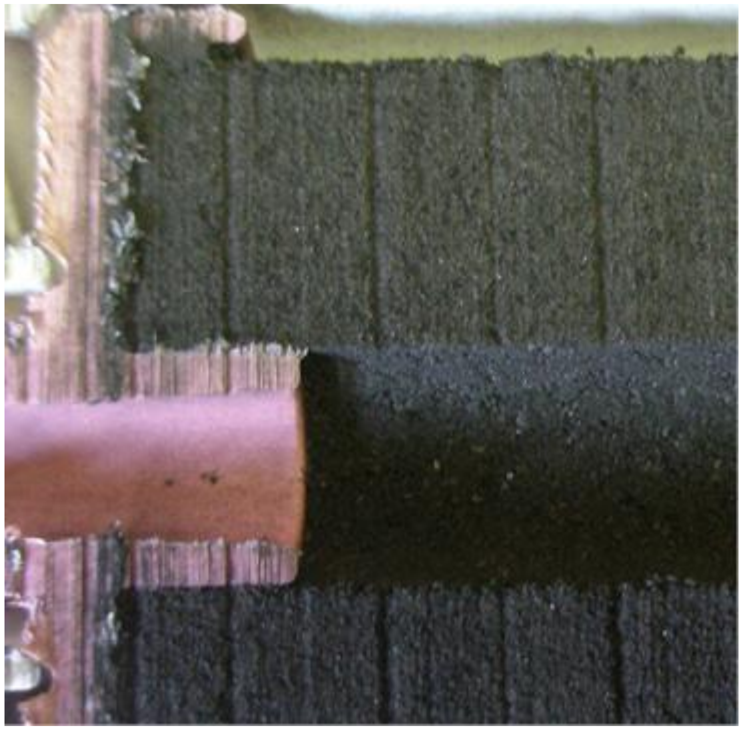


Figure 3.9

Activated carbon can be made from coal, wood or coconut shells. Carbon from coconut shells is the most expensive and most effective form. Carbon is “activated” by adding a positive charge, which enhances the adsorption of contaminants that have a negative charge. Some manufacturers use various blends of carbon to achieve specific water quality and contaminant reduction.

Three forms of activated carbon used in water filtration systems are granulated activated carbon (GAC), activated-carbon block and catalytic carbon. Catalytic carbon is an advanced activated-carbon product designed to adsorb chloramines. Chloramines replace chlorine in the disinfection process and have been found to form trihalomethanes (THMs), a cancer-causing substance. Catalytic carbons also may remove hydrogen sulfide gas, which produces the “rotten egg” smell in some well water.

While all are effective, activated carbon block filters generally remove more contaminants because of their greater surface area. The efficiency of these filters is determined by the length of time the contaminants are in contact with the carbon. The lower the flow rate of the water, the more time the contaminants will be in contact with the carbon and the more adsorption will take place.

Removal rates also are affected by the particle size. Activated-carbon filters usually are rated by the size of the particles they are able to remove, measured in microns, and generally range from 50 microns (least effective) to 0.5 micron (most effective). Before purchasing the unit, ask the dealer if the filter can be replaced, how often it should be replaced, where you can purchase replacement filters and how much they cost. Keep in mind that the suggested life of the carbon filter is an estimate, so you should pay attention to any reduction in water pressure at the water tap that might indicate the filter is saturated with particles.

Fig 3.10. Sediment and Carbon chamber front view

CHAPTER 4

TESTS

4.1 Sampling and testing

For regular 30 days, greywater sample was collected from the outlet of the apartment's pipeline and the characteristics of that raw greywater was tested for the further design of set-up i.e., its different treatment units for filtration so that it can be checked that the treated greywater by the different filtration unit can be considered for further reuse option or not. The tests conducted on the samples are chemical oxygen demand (COD), total dissolved solids (TDS), dissolved oxygen (DO), pH, turbidity, hardness, alkalinity, conductivity

4.2 Laboratory Scale Study for Reuse of Greywater

Analytical method used in laboratory

All the samples of greywater which were being collected as per sampling program were analyzed at the college laboratory of Buddha institute of technology, Gorakhpur. The important parameter like pH, turbidity, conductivity, alkalinity, hardness, DO, total dissolved solids, COD of the influent and effluent of the greywater were analyzed.

4.2.1 Test for pH

pH is measured by using digital pH meter PM 100.

4.2.2 Tested for Turbidity

Turbidity is measured by using Nephelometric Turbidity Meter.

4.2.3 Tested for Alkalinity

Alkalinity is measured by using APHA (American Paint Horse Association) standards.

4.2.4 Tested for Hardness

Hardness is measured by APHA (American Paint Horse Association) standards.

4.2.5 Total Dissolved Solid (TSS)

The method is to measure the amount of solids present in greywater sample in dissolved form by using wattman filter paper.

4.2.6 Tested for Dissolved Oxygen (DO)

Dissolved oxygen is measured by using APHA (American Paint Horse Association) standard.

CHAPTER 5

RESULTS & DISCUSSION

1- Effluent Quality: Present the results of water quality analysis before and after treatment using the bio-filter. Highlight the reduction in key parameters such as biochemical oxygen demand (BOD), total suspended solids (TSS), and fecal coliform counts, proving the effectiveness of the bio-filter in improving water quality.

Pathogen Removal: Discuss the removal efficiency of pathogens such as bacteria, viruses, and protozoa achieved by the bio-filter. Present the percentage reduction in pathogen concentrations before and after treatment, emphasizing the importance of microbial safety in recycled greywater for household use.

2- Operational Performance: Evaluate the operational performance of the bio-filter system, including factors such as hydraulic retention time, flow rates, and treatment efficiency. Present data on the system's stability and reliability over time, proving its suitability for continuous operation in rural household settings.

3- Cost Analysis: Provide a detailed cost analysis of the bio-filter system, including materials, construction, and maintenance expenses. Compare the total cost of implementation with alternative greywater treatment technologies, highlighting the cost-effectiveness of the bio-filter design.

4- User Satisfaction: Present feedback from end-users about their satisfaction with the bio-filter system. Discuss any challenges met during implementation and operation, as well as suggestions for improvement based on user experience and preferences.

RESULTS OF SAMPLE OF GREY WATER COLLECTED AFTER DOMESTIC USES AFTER TESTING FROM DISTRICT WATER TESTING LABORATORY (GORAKHPUR)-

Sr. No	Analyzed Parameters, (unit of measurement)	Observed Value	Unit	Acceptable Limit	Permissible Limit	Ref Method of Analysis
1	**Turbidity**	**15.8**	**NTU**	**1**	**5**	**IS 3025(Part 10):2017**
2	**pH (25°)**	**8.29**	**----**	**6.5-8.5**	**No** relaxation	**IS 3025(Part 11):2017**
3	**TDS**	**846**	**mg/L**	**500**	**2000**	**IS 3025(Part 16):2022**
4	**Total Hardness**	**246**	**mg/L**	**200**	**600**	**IS 3025(Part 21):2019**
5	**Total alkalinity**	**242**	**mg/L**	**200**	**600**	**IS 3025(Part 23):2019**
6	**Nitrate**	**1.465**	**mg/L**	**45**	**No** relaxation	**ALPHA 4500-NO_3 UV screening method 2017**
7	**Iron**	**1.126**	**mg/L**	**1**	**No** relaxation	**ALPHA 4500-NO_3 phenanthroline method 2017**

RESULTS OF SAMPLE OF GREY WATER COLLECTED AFTER DOMESTIC USES AFTER TESTING FROM DISTRICT WATER TESTING LABORATORY (GORAKHPUR)-

Sr. No	Analyzed Parameters,(unit of measurement)	Observed Value	Unit	Acceptable Limit	Permissible Limit	Ref Method of Analysis
1	**Turbidity**	**1.58**	**NTU**	**1**	**5**	**IS 3025(Part 10):2017**
2	**pH (25°)**	**7.8**	**----**	**6.5-8.5**	**No** relaxation	**IS 3025(Part 11):2017**
3	**TDS**	**641**	**Mg/L**	**500**	**2000**	**IS 3025(Part 16):2022**
4	**Total Hardness**	**224**	**Mg/L**	**200**	**600**	**IS 3025(Part 21):2019**
5	**Total alkalinity**	**124**	**Mg/L**	**200**	**600**	**IS 3025(Part 23):2019**
6	**Nitrate**	**0.862**	**Mg/L**	**45**	**No** relaxation	**ALPHA 4500-NO_3 UV screening method 2017**
7	**Iron**	**0.228**	**Mg/L**	**1**	**No** relaxation	**ALPHA 4500-**

Discussion-

As per the discussion with Mr. Ankur Kumar sir who is Research director of Buddha Group Institutions, he suggested us to design a low-cost portable bio-filter system that has the capability to treat grey water as well as it is capable to be used commercially at such places where a lot of amounts of grey water is generated and been thrown away as wastewater.

For the sake of use of bio filter commercially we meet a laundry shop owner in Khalilabad Mr. Vishal Ojha and Mr. Rudra Tripathi, they said that the daily use of water in their shop is approx. 3000 liters and their 70% of waste water is grey, they are happy to install such a bio filter which help them to reuse the waste water which eventually helps in the reduction of water demand for their shop and to put forward a step in the sustainable development.

1. Effectiveness of Bio-Filtration: Discuss the effectiveness of bio-filtration in treating greywater. Highlight how the design of the low-cost bio-filter eases the removal of contaminants and pathogens from greywater, making it suitable for reuse in rural households.
2. Cost-Effectiveness: Evaluate the cost-effectiveness of the designed bio-filter compared to traditional greywater treatment systems. Discuss how the use of locally available materials and simple construction techniques contributes to its affordability, making it suitable for implementation in rural areas with limited resources.
3. Sustainability: Explore the sustainability aspects of the bio-filter design. Discuss its potential to reduce the environmental impact of greywater

disposal by promoting recycling and reuse, thereby conserving water resources and reducing pollution in rural communities.

4. Scalability and Adaptability: Address the scalability and adaptability of the bio-filter design. Discuss how it can be scaled up or down to accommodate varying greywater volumes and quality levels, as well as its suitability for different geographical and socio-economic contexts.
5. Community Engagement: Highlight the importance of community engagement in the implementation of the bio-filter technology. Discuss the need for community participation in the design, construction, and maintenance of the bio-filters to ensure their long-term sustainability and effectiveness.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

1. The design and implementation of a low-cost bio-filter for greywater treatment hold immense promise for addressing water scarcity challenges and promoting sustainable water management practices in rural households. Throughout this research, we have proved the effectiveness of the bio-filter system in treating greywater, with the goal of enabling recycling and reuse within rural communities.

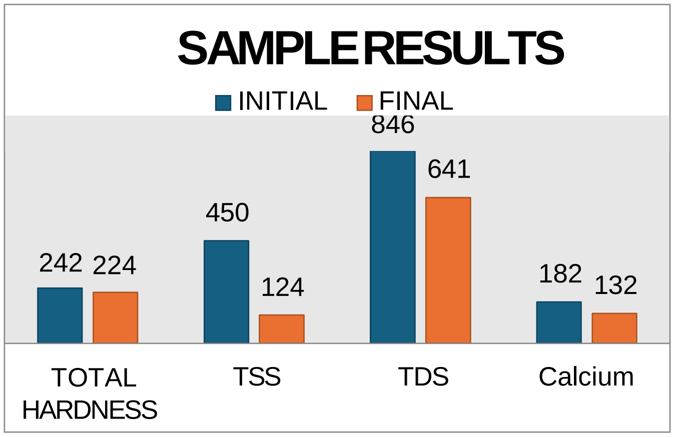
2. Our findings underscore the significance of decentralized greywater treatment technologies tailored to the specific needs and constraints of rural contexts. By harnessing natural processes and using locally available materials, the bio-filter system offers a practical and affordable solution for enhancing water quality and conserving precious water resources.

3. The successful deployment of the bio-filter system relies not only on technical efficacy but also on community engagement, user acceptance, and supportive policy frameworks. Moving forward, it is essential to prioritize efforts aimed at building ability, fostering collaboration, and advocating for policy changes to ease the widespread adoption of sustainable water reuse practices in rural areas.

4. In conclusion, this research is a step towards realizing the vision of water security, environmental sustainability, and socio-economic development in

rural communities. By using innovative technologies and embracing a comprehensive approach to water management, we can empower individuals and communities to thrive in harmony with their natural surroundings, ensuring a brighter and more resilient future for generations to come.

1. GRAPHICAL AND PICTORICAL DEMOSTRATION OF RESULTS



GRAPH 6.1&6.2

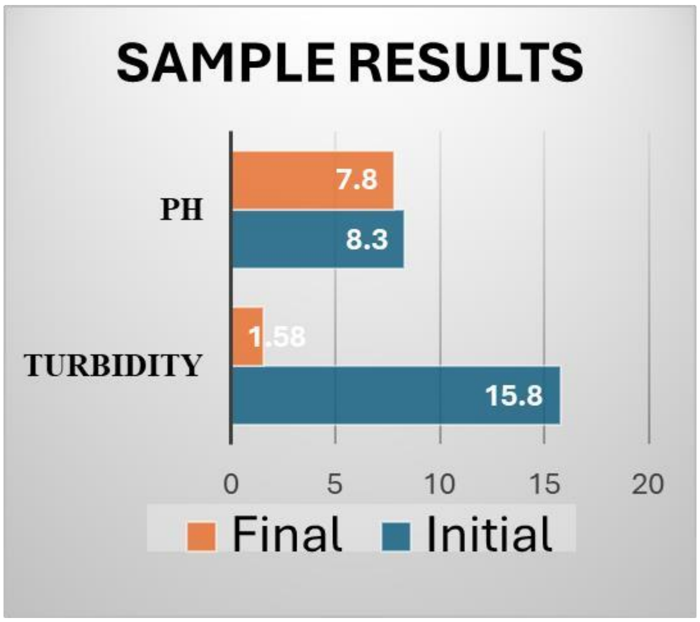


Fig 6.1. RAW SAMPLE OF TDS CALCULATION

Fig 5\6.2. TREATED SAMPLE OF TDS CALCULATION

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ACHIEVEMENTS

"WINNER OF **8TH TECHYUVA 2023** ORGANIZED BY BUDDHA GROUP OF INSTITUTIONS WITH COLLABORATION OF AKTU INNOVATION HUB AND IEEE YOUNG PROFESSIONALS UTTAR PRADESH".

PAPER PUBLICATION IN "JOURNAL OF WATER RESOURCE RESEARCH AND DEVELOPMENT" ACCEPTED AND IS IN UNDER PROCESS.